

Assignment: Autonomous AI Agents for Data Analysis and Visualization

Objective

This assignment challenges you to build a system of two distinct AI agents that work in tandem to analyze and visualize data. The primary goal is to create a scalable and user-friendly solution that can be controlled via a simple user interface (UI) or a command-line interface (CLI).

Project Description

You will develop two agents :

1. **Data Analysis Agent:** This agent will be responsible for processing and analyzing a given dataset. It should be designed to be controllable, allowing users to specify the analysis they require.
2. **Data Visualization Agent:** This agent will take the analyzed data and generate plots and charts in a predefined "company style" format.

The dataset for this assignment can be downloaded from the following link: [Dataset Link](#). You are encouraged to use existing agent frameworks. The two agents need to be connected.

Key Requirements

Your implementation must meet the following criteria:

- ☐ **User Controllability:** Both agents must be controllable through a simple UI or CLI.
- ☐ **User Restriction:** The ability to restrict specific users to certain databases.
- ☐ **Easy Integration:** A straightforward process for adding new databases.
- ☐ **Automated Testing:** You must provide at least three evaluation examples in the form of automated tests to demonstrate the functionality and correctness of your agents.

We will provide access to an Openai api key. Please ask for this if we have not yet provided it.

Deliverables

- ☐ Source code for the project
- ☐ 3-5 slide presentation (overview and design decisions)
- ☐ Automated tests with at least three evaluation examples.
- ☐ Live demonstration